Components

	f	V_{pp}
LF drift	600 Hz	0.8
Useful signal	6 kHz	0.4
Structured Interference	18 kHz	1.2
HF interference	55 kHz	2
Noise	broadband	150 mV rms

Stage 1: Buffer : (loading effect)

Load (filter) draws power from sensor.

Input impedance of filter module
$\gg$ output impedance of sensor

Op-amp has $\uparrow$ input impedance
$\downarrow$ output impedance

Stage 2:
To remove 18 kHz & 55 kHz
→ Low pass

Stage 3:
To remove 600 Hz → High pass filter
as it is below desired frequency

LPF + HPF → BPF centred
around 6 kHz

Limit bandwidth to improve signal
to noise ratio

2nd order filters preferred
as interference is much stronger
than desired
(1.2 Vpp, 2 Vpp) > 0.4 Vpp

gives −40 dB/decade roll off

Black box

↓

Input buffer — Non inverting op amp follows

↓

2nd order LPF — $f_c = 10\,kHz$

↓

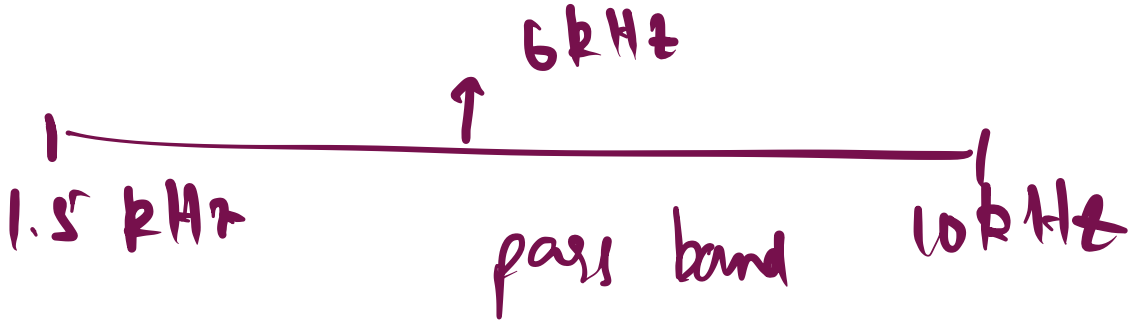
2nd order HPF — $f_c = 1.5\,kHz$

↓

Variable Gain Amplifier

↓

output



Sallen-key active filter

→ stable
→ smooth response

White noise ∝ bandwidth

Designers should first limit bandwidth and then analyse other requirements like gain.

Potentiometre → Voltage gain control

Design

i) LPF, $f_c = 10\ kHz$

$C = 10\ nF = C_2 = C_3$

$$f_H = \frac{1}{2\pi\sqrt{R_2 R_3 C_2 C_3}} = \frac{1}{2\pi RC}$$

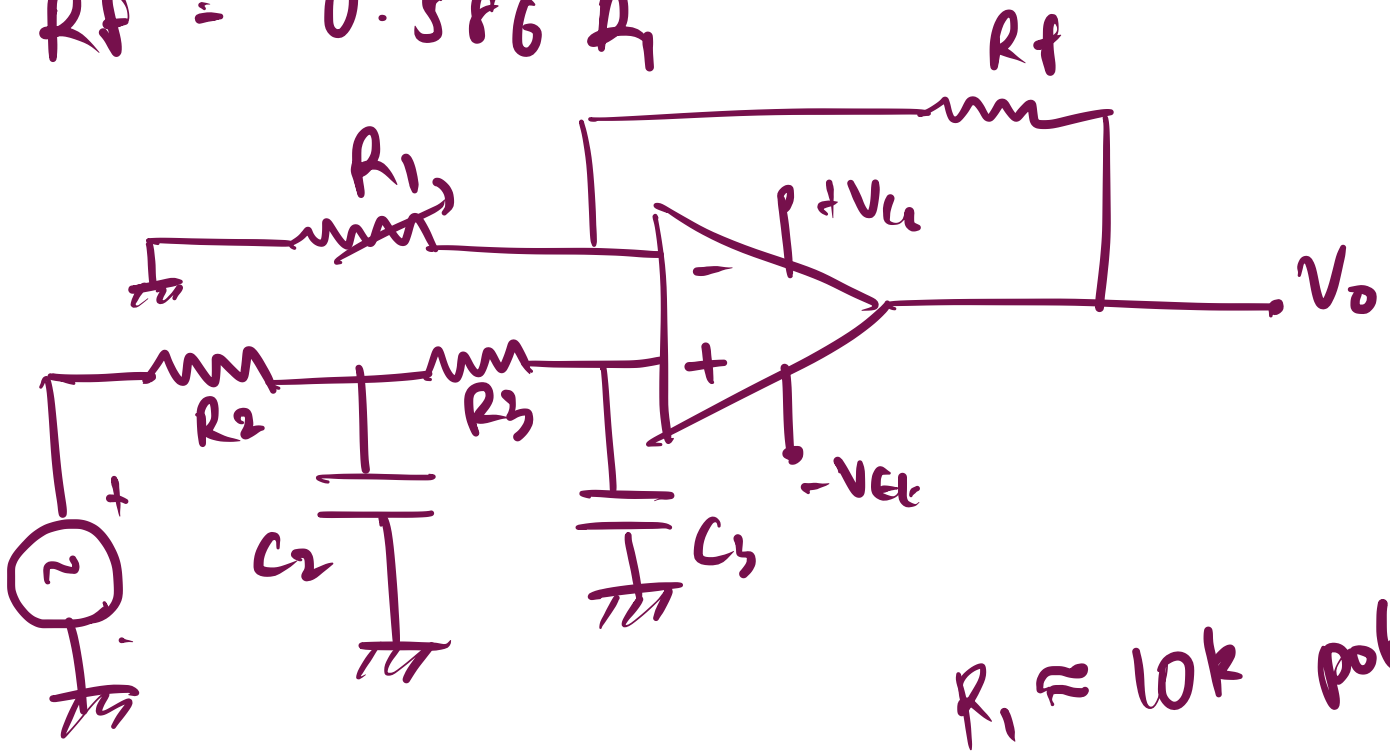
for $R = R_2 = R_3$

$$R = \frac{1}{2\pi f C}$$

$$= \frac{1}{2\pi \times 10\times 10^3 \times 10\times 10^{-9}}$$

$$= \frac{1}{2\pi \times 10^{-4}} = 1591.54\ \Omega \approx 1.6\ k\Omega$$

$R_f = 0.586\ R_1$



$R_1 \approx 10k$ pot

HPF

$f_c = 1.5\ kHz$

$R = 10\ k\Omega$

$$f_c = \frac{1}{2\pi RC}$$

$$C = \frac{1}{2\pi f_c R}$$

$$= \frac{1}{2\pi \times 1.5 \times 10^3 \times 10 \times 10^3}$$

$$= 1.061 \times 10^{-8}\ F$$

$$= 10.61\ nF$$

$$\frac{R_f}{R_i} = 1.585$$

$$R_f = 15.85\ k\Omega$$

$f_{cen} = 6\ kHz$

$f_{BW} = f_H - f_L$

$= 9\ kHz - 3\ kHz$

$= 6\ kHz$

$Q = 1$

$C = 10\ nF$

$R_3 = \frac{2Q}{\omega_{cen} C} = \frac{2Q}{2\pi f_{cen} C}$

$= \frac{2(1)}{2\pi \times 6 \times 10^3 \times 10 \times 10^{-9}}$

$= \frac{1}{\pi \times 6 \times 10^4 \times 10^{-9}}$

$= \frac{1}{\ }$

$= 5305.1647\ \Omega$

$R_2 = \frac{R_3}{4Q^2 - 2K} = \frac{5305.1647}{4 - 2(1)}$

$= 331.57\ \Omega$

$R_1 = \frac{R_3}{2K} = 265.858$